Coverage Metric - GPU Optimized

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```
# Configuración para visualización en PDF
import pandas as pd
import numpy as np
import sys

# Para que el texto de pandas no se corte
pd.set_option('display.max_colwidth', None)
pd.set_option('display.max_columns', None)
pd.set_option('display.width', 100)  # Ancho fijo para pandas

# Para prints largos de numpy - ancho más conservador
np.set_printoptions(linewidth=100, edgeitems=3)

# Configurar ancho de terminal para prints
import os
os.environ['COLUMNS'] = '100'

print(" Configuración para PDF lista")
```

```
 Configuración para PDF lista
```

0.1 1. Setup e Imports

```
import numpy as np
import torch
import sys
import os
from pathlib import Path
from tqdm import tqdm
import time

# Configurar paths
project_root = Path('../../..').resolve()
sys.path.insert(0, str(project_root))
```

```
from sae.models.sae import SparseAutoencoder
from sae.tools.naming_utils import get_checkpoint_dir, get_model_name, get_extras_id
from sae.tools.experiment_utils import get_experiment_dir, get_metrics_dir, get_report_name

# Verificar GPU
device = torch.device('cuda' if torch.cuda.is_available() else 'cpu')
print(f"Device: {device}")
if torch.cuda.is_available():
    print(f"GPU: {torch.cuda.get_device_name(0)}")
    print(f"Memoria disponible: {torch.cuda.get_device_properties(0).total_memory / 1e9:.2f} GB")
```

```
Device: cuda
GPU: NVIDIA GeForce RTX 5060
Memoria disponible: 8.55 GB
```

```
# Metadata para naming de checkpoints (separada de hiperparámetros)
NAMING_META = {
    'variante': 'standard',          # Arquitectura del SAE
    'tecnica': 'l1-decay',                 # L1 con penalty annealing
    'capa': 6,                       # Layer del modelo OthelloGPT
    'juegos': 500,                  # Número de partidas
    'base_path': 'C:\\Users\\Esposa\\Documents\\Repos\\sae-othello-gpt',  # Path base para checkpoints
    'experiment_base_path': os.path.abspath('../../experiments') ,      # sae/experiments/
    'extras': {
        'expansion': 32,
        'sparsity_coef': 5e-4,
        'epochs': 200,
        'patience': 15,
        'learning_rate': 1e-3,
        'sparsity_decay': 0.95,
        'batch_size': 256
    }
}
print('\n Metadata del experimento:')
for key, value in NAMING_META.items():
    print(f'  {key}: {value}')
```

```
 Metadata del experimento:
  variante: standard
  tecnica: l1-decay
  capa: 6
  juegos: 500
  base_path: C:\Users\Esposa\Documents\Repos\sae-othello-gpt
  experiment_base_path: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments
  extras: {'expansion': 32, 'sparsity_coef': 0.0005, 'epochs': 200, 'patience': 15, 'learning_rate': 0.001, 'sparsity_d
```

0.2 2. Cargar Datos

```
# Cargar activaciones pre-SAE
activations_path = project_root / "sae" / "activations" / "data" / "layer6_200games.npy"
activations_full = np.load(activations_path)
####
activations = activations_full

print(f"Activaciones del modelo:")
print(f"  Shape original: {activations_full.shape}")
print(f"  Shape final: {activations.shape}")
print(f"  Memoria: {activations.nbytes / (1024**2):.2f} MB")
```

```
Activaciones del modelo:
  Shape original: (11800, 512)
  Shape final: (11800, 512)
  Memoria: 23.05 MB
```

```python
# Cargar ground truth de BSPs
bsp_gt_path = project_root / "sae" / "metrics" / "02_data" / "bsp_ground_truth_200games.npy"
bsp_names_path = project_root / "sae" / "metrics" / "02_data" / "bsp_ground_truth_200games.names.npy"

bsp_ground_truth_full = np.load(bsp_gt_path)
bsp_names = np.load(bsp_names_path, allow_pickle=True)
###
indices = np.random.choice(len(activations_full), size=1000, replace=False)
activations = activations_full[indices]
##
bsp_ground_truth = bsp_ground_truth_full[indices]
print(f"\nGround truth BSPs:")
print(f"  Shape original: {bsp_ground_truth_full.shape}")
print(f"  Shape final: {bsp_ground_truth.shape}")
print(f"  Total BSPs: {len(bsp_names)}")
```

```
Ground truth BSPs:
  Shape original: (11800, 326)
  Shape final: (1000, 326)
  Total BSPs: 326
```

0.3 3. Cargar SAE y Extraer Features

```python
# Configuración del SAE
input_dim = 512
hidden_dim = 16384

# Construir ruta del modelo usando naming_utils
checkpoint_dir = get_checkpoint_dir(
    NAMING_META['base_path'],
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos']
)
extras_id = get_extras_id(checkpoint_dir, NAMING_META['extras']) if NAMING_META.get('extras') else None

model_name = get_model_name(
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos'],
    'best',
    extras_id=extras_id
)
model_path = checkpoint_dir / model_name

# Cargar modelo (weights_only=False: checkpoint contiene objetos Path en config)
sae = SparseAutoencoder(input_dim, hidden_dim).to(device)
checkpoint = torch.load(model_path, map_location=device, weights_only=False)
sae.load_state_dict(checkpoint['model_state_dict'])
sae.eval()

print(f"SAE cargado:")
```

```
print(f"  Path: {model_path}")
print(f"  Input: {input_dim}, Hidden: {hidden_dim}")
print(f"  Expansion: {hidden_dim/input_dim}x")
print(f"  Epoch: {checkpoint.get('epoch', 'N/A')}, Val MSE: {checkpoint.get('val_mse', float('nan')):.6f}")
```

```
SAE cargado:
  Path: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\sae_l1-decay_standard\500games\sae
decay_l6_500g_ex6_best.pt
  Input: 512, Hidden: 16384
  Expansion: 32.0x
  Epoch: N/A, Val MSE: nan
```

```
print("Extrayendo features del SAE...")
activations_tensor = torch.from_numpy(activations).float().to(device)

# PRIMERO: calcular sae_features con los 15 tableros
with torch.no_grad():
    sae_features = torch.relu(sae.encoder(activations_tensor))

# SEGUNDO: calcular f_max_global sobre TODO el dataset
activations_full_tensor = torch.from_numpy(activations_full).float().to(device)
with torch.no_grad():
    sae_features_full = torch.relu(sae.encoder(activations_full_tensor))
f_max_global = sae_features_full.max(dim=0)[0]

# Liberar memoria
del activations_full_tensor, sae_features_full
torch.cuda.empty_cache()
print("f_max global calculado sobre todo el dataset")

print(f"\nFeatures SAE:")
print(f"  Shape: {sae_features.shape}")
print(f"  Device: {sae_features.device}")
print(f"  Sparsity: {(sae_features == 0).float().mean():.2%}")
print(f"  Activaciones promedio: {(sae_features > 0).sum(dim=1).float().mean():.1f}")
```

```
Extrayendo features del SAE...
f_max global calculado sobre todo el dataset

Features SAE:
  Shape: torch.Size([1000, 16384])
  Device: cuda:0
  Sparsity: 95.45%
  Activaciones promedio: 745.1
```

0.4 4. Filtrar BSPs de Piezas

Coverage solo usa las 128 BSPs de piezas (mías/oponente), sin vacías.

```
# ============================================
# CONFIGURACIÓN DE FILTRADO DE BSPs
# ============================================
USE_PLAYER   = False   # BSPs perspectiva del jugador (1 y 2)
USE_ABSOLUTE = True  # BSPs absolutas negro/blanco (B y W)
USE_STRATEGIC = False # BSPs especiales (BSP_)

# Filtrar según configuración
bsp_pieces_indices = []
bsp_pieces_names = []
```

```
for i, name in enumerate(bsp_names):
    include = False

    if USE_PLAYER and len(name) == 6 and name.startswith('BSP') and (name.endswith('1') or name.endswith('2')):
        include = True
    if USE_ABSOLUTE and (name.endswith('B') or name.endswith('W')):
        include = True
    if USE_STRATEGIC and name.startswith('BSP_'):
        include = True

    if include:
        bsp_pieces_indices.append(i)
        bsp_pieces_names.append(name)

bsp_pieces_indices = np.array(bsp_pieces_indices)
bsp_pieces_gt = bsp_ground_truth[:, bsp_pieces_indices]

# Convertir a tensor en GPU
bsp_pieces_gt_tensor = torch.from_numpy(bsp_pieces_gt).bool().to(device)

print(f"BSPs seleccionadas para Coverage:")
print(f"  Player: {USE_PLAYER} | Absolute: {USE_ABSOLUTE} | Strategic: {USE_STRATEGIC}")
print(f"  Total: {len(bsp_pieces_indices)}")
print(f"  Shape: {bsp_pieces_gt_tensor.shape}")
print(f"  Device: {bsp_pieces_gt_tensor.device}")
print(f"  Primeras 5: {', '.join(bsp_pieces_names[:5])}")
print(f"  Últimas 5: {', '.join(bsp_pieces_names[-5:])}")
# Identificador del filtrado (para naming de archivos de salida)
parts = []
if USE_PLAYER: parts.append("player")
if USE_ABSOLUTE: parts.append("absolute")
if USE_STRATEGIC: parts.append("strategic")
filter_suffix = "_".join(parts) if parts else "none"
print(f"  Filter suffix: {filter_suffix}")
```

```
BSPs seleccionadas para Coverage:
  Player: False | Absolute: True | Strategic: False
  Total: 128
  Shape: torch.Size([1000, 128])
  Device: cuda:0
  Primeras 5: BSPA1B, BSPA1W, BSPA2B, BSPA2W, BSPA3B
  Últimas 5: BSPH6W, BSPH7B, BSPH7W, BSPH8B, BSPH8W
  Filter suffix: absolute
```

0.5 5. Implementación de Coverage

0.5.1 Estrategia de optimización:

1. **Vectorización**: Procesar múltiples features en paralelo usando operaciones matriciales
2. **Batch processing**: Dividir en chunks para no saturar memoria GPU
3. **Pre-cálculo**: Calcular máximos una sola vez
4. **Early stopping**: Terminar cuando F1 1.0

```
def fast_f1_score_gpu(y_true, y_pred, eps=1e-8):
    """
    F1 score vectorizado en GPU.

    Args:
        y_true: Tensor (n_positions,) booleano
        y_pred: Tensor (n_positions, n_candidates) booleano
```

```
    Returns:
        f1_scores: Tensor (n_candidates,)
    """
    # y_true: (n_positions,) -> expand to (n_positions, 1)
    y_true_expanded = y_true.unsqueeze(1)  # (n_positions, 1)

    # Calcular TP, FP, FN
    tp = (y_true_expanded & y_pred).sum(dim=0).float()  # (n_candidates,)
    fp = (~y_true_expanded & y_pred).sum(dim=0).float()
    fn = (y_true_expanded & ~y_pred).sum(dim=0).float()

    # Precision y Recall
    precision = tp / (tp + fp + eps)
    recall = tp / (tp + fn + eps)

    # F1
    f1 = 2 * (precision * recall) / (precision + recall + eps)

    return f1


def calculate_coverage_gpu_optimized(
    sae_features,  # Tensor (n_positions, n_features) en GPU
    bsp_ground_truth,  # Tensor (n_positions, n_bsps) en GPU, bool
    thresholds=[0.0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9],
    batch_size_features=512,  # Procesar 512 features a la vez
    f_max=None,  # ← AGREGAR
    verbose=True
):
    """
    Calcula Coverage de forma optimizada para GPU.

    Procesa múltiples features y thresholds en paralelo usando operaciones matriciales.
    """
    n_positions, n_features = sae_features.shape
    n_bsps = bsp_ground_truth.shape[1]
    n_thresholds = len(thresholds)

    if verbose:
        print("="*60)
        print("CALCULATING COVERAGE (GPU OPTIMIZED)")
        print("="*60)
        print(f"Positions: {n_positions:,}")
        print(f"SAE Features: {n_features:,}")
        print(f"BSPs: {n_bsps}")
        print(f"Thresholds: {n_thresholds}")
        print(f"Batch size (features): {batch_size_features}")
        print(f"Total evaluations: {n_bsps * n_features * n_thresholds:,}")
        print("="*60)

    # Pre-calcular máximos de features
    if f_max is None:
        f_max = sae_features.max(dim=0)[0]
    active_features = (f_max > 0).nonzero(as_tuple=True)[0]
    n_active = len(active_features)

    if verbose:
        print(f"\nActive features: {n_active:,}/{n_features:,} ({n_active/n_features*100:.1f}%)")
        print(f"Reduced evaluations: {n_bsps * n_active * n_thresholds:,}")
        print()

    # Almacenar resultados
```

```
    best_f1s = torch.zeros(n_bsps, device=sae_features.device)
    best_features = torch.full((n_bsps,), -1, dtype=torch.long, device=sae_features.device)
    best_thresholds = torch.full((n_bsps,), -1.0, device=sae_features.device)

    thresholds_tensor = torch.tensor(thresholds, device=sae_features.device)

    # Procesar cada BSP
    pbar = tqdm(range(n_bsps), desc="Processing BSPs") if verbose else range(n_bsps)

    for bsp_idx in pbar:
        bsp_labels = bsp_ground_truth[:, bsp_idx]  # (n_positions,)

        # Skip si la BSP nunca está activa
        if not bsp_labels.any():
            continue

        # Procesar features activas en batches
        for batch_start in range(0, n_active, batch_size_features):
            batch_end = min(batch_start + batch_size_features, n_active)
            batch_features_idx = active_features[batch_start:batch_end]

            # Extraer features del batch
            batch_activations = sae_features[:, batch_features_idx]  # (n_positions, batch_size)
            batch_f_max = f_max[batch_features_idx]  # (batch_size,)

            # Para cada threshold, binarizar TODAS las features del batch a la vez
            for t in thresholds_tensor:
                # Binarizar: (n_positions, batch_size)
                predictions = batch_activations > (t * batch_f_max.unsqueeze(0))

                # Calcular F1 para todas las features del batch
                f1_scores = fast_f1_score_gpu(bsp_labels, predictions)  # (batch_size,)

                # Encontrar la mejor en este batch
                max_f1, max_idx_in_batch = f1_scores.max(dim=0)

                # Actualizar si es mejor que lo visto hasta ahora
                if max_f1 > best_f1s[bsp_idx]:
                    best_f1s[bsp_idx] = max_f1
                    best_features[bsp_idx] = batch_features_idx[max_idx_in_batch]
                    best_thresholds[bsp_idx] = t

            # Early stopping: si ya es casi perfecto, no revisar más batches
            if best_f1s[bsp_idx] >= 0.999:
                break

    # Calcular Coverage (macro-average)
    coverage = best_f1s.mean().item()

    # Convertir a CPU para retornar
    best_f1s_cpu = best_f1s.cpu().numpy()
    best_features_cpu = best_features.cpu().numpy()
    best_thresholds_cpu = best_thresholds.cpu().numpy()

    return coverage, best_f1s_cpu, best_features_cpu, best_thresholds_cpu

print(" Funciones de Coverage GPU definidas")
```

```
 Funciones de Coverage GPU definidas
```

0.6 6. Calcular Coverage

```
# Medir tiempo de ejecución
start_time = time.time()

coverage, best_f1s, best_features, best_thresholds = calculate_coverage_gpu_optimized(
    sae_features,
    bsp_pieces_gt_tensor,
    batch_size_features=512,  # Ajustar según memoria GPU
    f_max=f_max_global,
    verbose=True
)

elapsed_time = time.time() - start_time

print("\n" + "="*60)
print("RESULTADO COVERAGE")
print("="*60)
print(f"Coverage Score: {coverage:.4f}")
print(f"Objetivo (paper): 0.52")
print(f"Diferencia: {coverage - 0.52:+.4f}")
print(f"\nTiempo de ejecución: {elapsed_time:.2f} seg")
print(f"                     ({elapsed_time/60:.2f} min)")
print("="*60)
```

```
============================================================
CALCULATING COVERAGE (GPU OPTIMIZED)
============================================================
Positions: 1,000
SAE Features: 16,384
BSPs: 128
Thresholds: 10
Batch size (features): 512
Total evaluations: 20,971,520
============================================================

Active features: 11,137/16,384 (68.0%)
Reduced evaluations: 14,255,360

Processing BSPs: 100%|      | 128/128 [00:19<00:00,  6.66it/s]


============================================================
RESULTADO COVERAGE
============================================================
Coverage Score: 0.4623
Objetivo (paper): 0.52
Diferencia: -0.0577

Tiempo de ejecución: 19.24 seg
                     (0.32 min)
============================================================
```

0.7 7. Análisis de Resultados

```python
# Estadísticas de distribución de F1 scores
print("Distribución de F1 scores:")
print(f"  Mínimo: {best_f1s.min():.4f}")
print(f"  Máximo: {best_f1s.max():.4f}")
print(f"  Media: {best_f1s.mean():.4f}")
print(f"  Mediana: {np.median(best_f1s):.4f}")
print(f"  Std: {best_f1s.std():.4f}")
print()
print(f"BSPs con F1 > 0.8: {(best_f1s > 0.8).sum()} ({(best_f1s > 0.8).mean()*100:.1f}%)")
print(f"BSPs con F1 > 0.5: {(best_f1s > 0.5).sum()} ({(best_f1s > 0.5).mean()*100:.1f}%)")
print(f"BSPs con F1 < 0.2: {(best_f1s < 0.2).sum()} ({(best_f1s < 0.2).mean()*100:.1f}%)")
```

```
Distribución de F1 scores:
  Mínimo: 0.3173
  Máximo: 0.7198
  Media: 0.4623
  Mediana: 0.4426
  Std: 0.0930

BSPs con F1 > 0.8: 0 (0.0%)
BSPs con F1 > 0.5: 38 (29.7%)
BSPs con F1 < 0.2: 0 (0.0%)
```

```python
# Top 10 BSPs mejor detectadas
top_indices = np.argsort(best_f1s)[-10:][::-1]

print("\nTop 10 BSPs mejor detectadas:")
print("="*60)
print(f"{'BSP':<10} {'F1':<8} {'Feature':<10} {'Threshold':<10}")
print("-"*60)
for idx in top_indices:
    bsp_name = bsp_pieces_names[idx]
    f1 = best_f1s[idx]
    feat = best_features[idx]
    thresh = best_thresholds[idx]
    print(f"{bsp_name:<10} {f1:<8.4f} {feat:<10} {thresh:<10.1f}")
```

```
Top 10 BSPs mejor detectadas:
============================================================
BSP        F1       Feature    Threshold
------------------------------------------------------------
BSPE4B     0.7198   10207      0.0
BSPD4B     0.6967   11324      0.0
BSPD5B     0.6852   10207      0.0
BSPE5B     0.6671   204        0.0
BSPE5W     0.6667   4255       0.0
BSPD3B     0.6657   11324      0.0
BSPD5W     0.6599   1947       0.1
BSPD4W     0.6387   204        0.0
BSPE4W     0.6239   4373       0.1
BSPC5W     0.6156   8880       0.1
```

```python
# Bottom 10 BSPs peor detectadas
bottom_indices = np.argsort(best_f1s)[:10]

print("\nBottom 10 BSPs peor detectadas:")
print("="*60)
print(f"{'BSP':<10} {'F1':<8} {'Feature':<10} {'Threshold':<10}")
print("-"*60)
for idx in bottom_indices:
```

```
    bsp_name = bsp_pieces_names[idx]
    f1 = best_f1s[idx]
    feat = best_features[idx]
    thresh = best_thresholds[idx]
    print(f"{bsp_name:<10} {f1:<8.4f} {feat:<10} {thresh:<10.1f}")
```

```
Bottom 10 BSPs peor detectadas:
============================================================
BSP        F1       Feature    Threshold
------------------------------------------------------------
BSPC8B     0.3173   8335       0.3
BSPF1W     0.3184   9630       0.2
BSPB8B     0.3189   10737      0.4
BSPG8B     0.3305   1947       0.3
BSPA7W     0.3382   11231      0.3
BSPF8B     0.3416   3941       0.3
BSPH8B     0.3426   9630       0.3
BSPA8B     0.3432   9630       0.4
BSPH7B     0.3453   9630       0.3
BSPH6B     0.3518   9630       0.2
```

```
# Distribución de thresholds óptimos
unique_thresholds, counts = np.unique(best_thresholds, return_counts=True)

print("\nDistribución de thresholds óptimos:")
print("="*60)
for t, count in zip(unique_thresholds, counts):
    if t >= 0:  # Ignorar -1 (BSPs sin match)
        percentage = count / len(best_thresholds) * 100
        print(f"Threshold {t:.1f}: {count:3d} BSPs ({percentage:.1f}%)")
```

```
Distribución de thresholds óptimos:
============================================================
Threshold 0.0:  11 BSPs (8.6%)
Threshold 0.1:  32 BSPs (25.0%)
Threshold 0.2:  57 BSPs (44.5%)
Threshold 0.3:  22 BSPs (17.2%)
Threshold 0.4:   6 BSPs (4.7%)
```

```
# ============================================
# CONSULTA POR BSP ESPECÍFICA
# ============================================

# >>> CAMBIA AQUÍ LA BSP QUE QUIERES ANALIZAR <
BSP_OBJETIVO = "BSPE4B"

# Buscar índice de la BSP
bsp_idx = bsp_pieces_names.index(BSP_OBJETIVO)

# Extraer resultados ya calculados
neurona   = best_features[bsp_idx]
threshold = best_thresholds[bsp_idx]
f1        = best_f1s[bsp_idx]

# f_max de la neurona ganadora (sae_features ya está en memoria)
f_max_neurona = sae_features[:, neurona].max().item()

# Activaciones de esa neurona en todos los tableros
activaciones_neurona = sae_features[:, neurona].cpu().numpy()
```

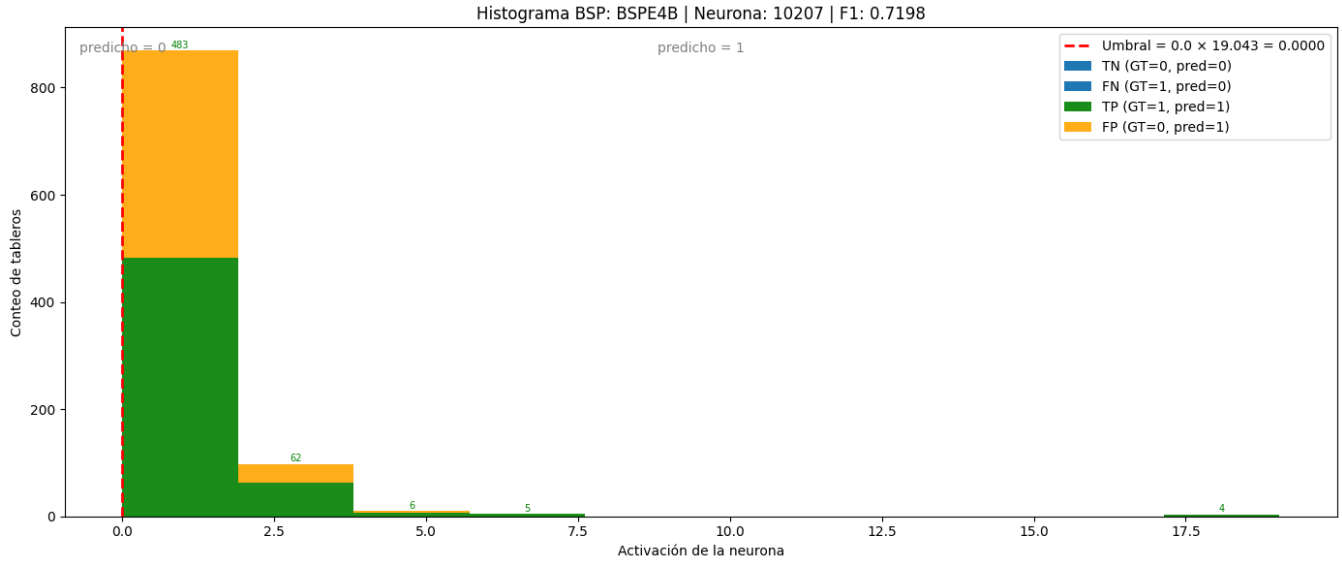
```
print(f"BSP analizada:     {BSP_OBJETIVO}")
print(f"Neurona ganadora:  {neurona}")
print(f"Threshold óptimo:  {threshold:.1f}")
print(f"F_max neurona:     {f_max_neurona:.4f}")
print(f"Umbral efectivo:   {threshold:.1f} x {f_max_neurona:.4f} = {threshold * f_max_neurona:.4f}")
print(f"F1 score:          {f1:.4f}")
print(f"\nActivaciones de neurona {neurona} ({len(activaciones_neurona)} tableros):")
for i, act in enumerate(activaciones_neurona):
    print(f"  Tablero {i+1:5d}: {act:.6f}")
```

```
def plot_bsp_histogram(BSP_HISTOGRAMA, sae_features, bsp_pieces_names, bsp_pieces_gt_tensor,
                       best_features, best_thresholds, best_f1s, n_bins=20):

    # Extraer datos de la BSP
    bsp_idx_hist   = bsp_pieces_names.index(BSP_HISTOGRAMA)
    neurona_hist   = best_features[bsp_idx_hist]
    threshold_hist = best_thresholds[bsp_idx_hist]
    f1_hist        = best_f1s[bsp_idx_hist]
    f_max_hist     = sae_features[:, neurona_hist].max().item()

    # Activaciones y ground truth
    activaciones = sae_features[:, neurona_hist].cpu().numpy()
    gt           = bsp_pieces_gt_tensor[:, bsp_idx_hist].cpu().numpy()
    umbral       = threshold_hist * f_max_hist

    # Clasificar
    predicho = (activaciones > umbral).astype(int)
    tp_mask  = (gt == 1) & (predicho == 1)
    fp_mask  = (gt == 0) & (predicho == 1)
    tn_mask  = (gt == 0) & (predicho == 0)
    fn_mask  = (gt == 1) & (predicho == 0)

    # Bins
    bins        = np.linspace(0, activaciones.max(), n_bins + 1)
    bin_centers = (bins[:-1] + bins[1:]) / 2
    bin_width   = bins[1] - bins[0]

    tp_counts, _ = np.histogram(activaciones[tp_mask], bins=bins)
    fp_counts, _ = np.histogram(activaciones[fp_mask], bins=bins)
    tn_counts, _ = np.histogram(activaciones[tn_mask], bins=bins)
    fn_counts, _ = np.histogram(activaciones[fn_mask], bins=bins)

    left_mask  = bin_centers <= umbral
    right_mask = bin_centers > umbral

    # Graficar
    fig, ax = plt.subplots(figsize=(14, 6))

    ax.bar(bin_centers[left_mask], tn_counts[left_mask],
           width=bin_width, color='steelblue', alpha=0.9,
           label='TN (GT=0, pred=0)', align='center')
    ax.bar(bin_centers[left_mask], fn_counts[left_mask],
           width=bin_width, bottom=tn_counts[left_mask],
           color='lightcoral', alpha=0.9,
           label='FN (GT=1, pred=0)', align='center')
    ax.bar(bin_centers[right_mask], tp_counts[right_mask],
           width=bin_width, color='green', alpha=0.9,
           label='TP (GT=1, pred=1)', align='center')
    ax.bar(bin_centers[right_mask], fp_counts[right_mask],
           width=bin_width, bottom=tp_counts[right_mask],
           color='orange', alpha=0.9,
```

```
            label='FP (GT=0, pred=1)', align='center')

    # Números TN
    for i, x in enumerate(bin_centers[left_mask]):
        total = tn_counts[left_mask][i] + fn_counts[left_mask][i]
        if tn_counts[left_mask][i] > 0:
            ax.text(x, total + 0.3, str(tn_counts[left_mask][i]),
                    ha='center', va='bottom', fontsize=7, color='steelblue')

    # Números TP
    for i, x in enumerate(bin_centers[right_mask]):
        total = tp_counts[right_mask][i] + fp_counts[right_mask][i]
        if tp_counts[right_mask][i] > 0:
            ax.text(x, total + 0.3, str(tp_counts[right_mask][i]),
                    ha='center', va='bottom', fontsize=7, color='green')

    # Umbral
    ax.axvline(x=umbral, color='red', linewidth=2, linestyle='--',
               label=f'Umbral = {threshold_hist:.1f} × {f_max_hist:.3f} = {umbral:.4f}')

    ax.set_xlabel('Activación de la neurona')
    ax.set_ylabel('Conteo de tableros')
    ax.set_title(f'Histograma BSP: {BSP_HISTOGRAMA} | Neurona: {neurona_hist} | F1: {f1_hist:.4f}')
    ax.legend()
    ax.text(umbral * 0.5, ax.get_ylim()[1] * 0.95, 'predicho = 0',
            ha='center', fontsize=10, color='gray')
    ax.text(umbral + (activaciones.max() - umbral) * 0.5, ax.get_ylim()[1] * 0.95, 'predicho = 1',
            ha='center', fontsize=10, color='gray')

    plt.tight_layout()
    plt.show()

    print(f"\nResumen:")
    print(f"  TP: {tp_mask.sum()} | FP: {fp_mask.sum()}")
    print(f"  TN: {tn_mask.sum()} | FN: {fn_mask.sum()}")
    print(f"  F1: {f1_hist:.4f}")

print("  Función plot_bsp_histogram definida")
```

```
  Función plot_bsp_histogram definida
```

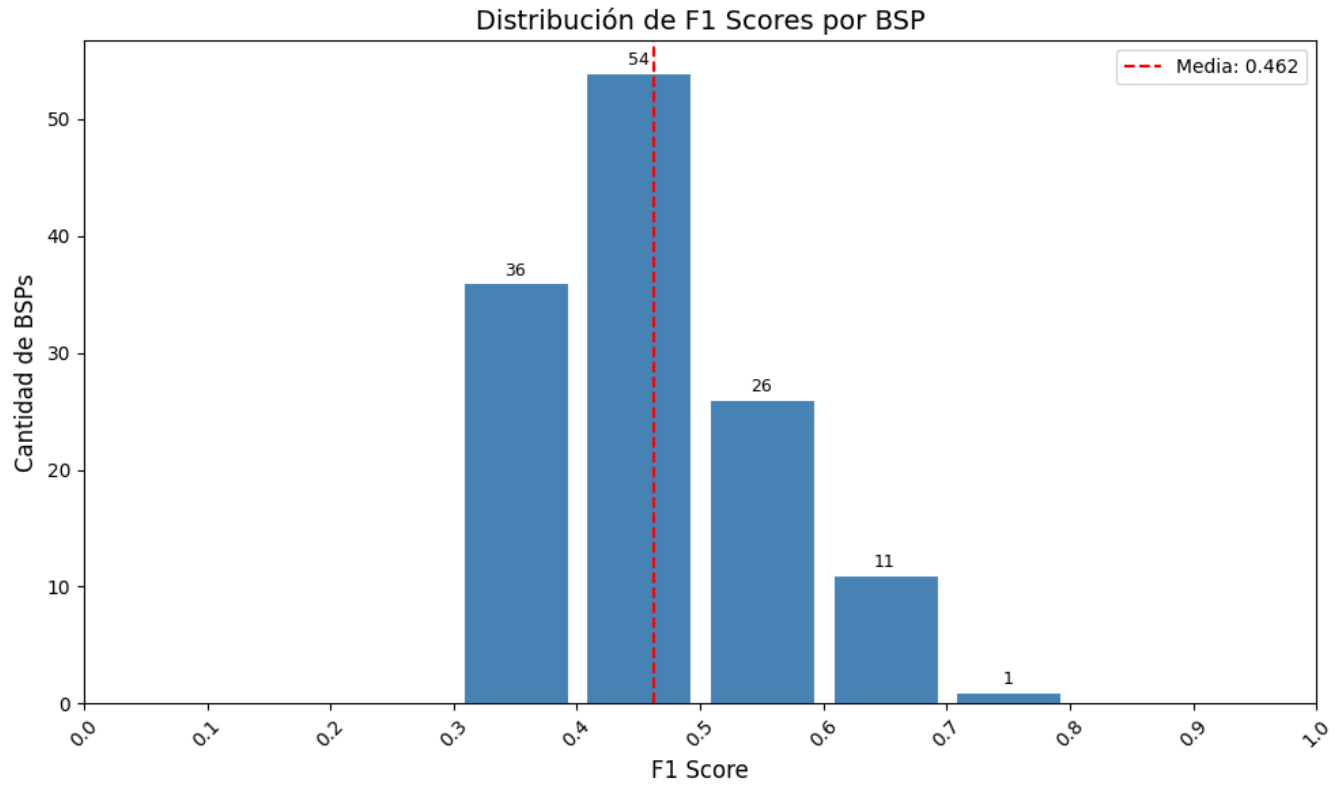
```
# >>> CAMBIA AQUÍ LA BSP QUE QUIERES VISUALIZAR <
import matplotlib.pyplot as plt
plot_bsp_histogram(
    BSP_HISTOGRAMA       = "BSPE4B",
    sae_features         = sae_features,
    bsp_pieces_names     = bsp_pieces_names,
    bsp_pieces_gt_tensor = bsp_pieces_gt_tensor,
    best_features        = best_features,
    best_thresholds      = best_thresholds,
    best_f1s             = best_f1s,
    n_bins               = 10
)
```



Resumen:
  TP: 560 | FP: 426
  TN: 4 | FN: 10
  F1: 0.7198

```python
import matplotlib.pyplot as plt
import numpy as np

def plot_f1_distribution(best_f1s, bsp_pieces_names, n_bins=10):
    """
    Gráfico 1: Distribución de F1 scores por BSP.

    Args:
        best_f1s: array numpy con el mejor F1 score de cada BSP
        bsp_pieces_names: lista con los nombres de las BSPs
        n_bins: número de bins del histograma
    """
    fig, ax = plt.subplots(figsize=(10, 6))

    bins = np.linspace(0, 1, n_bins + 1)
    counts, edges = np.histogram(best_f1s, bins=bins)
    bin_centers = (edges[:-1] + edges[1:]) / 2
    bin_width = edges[1] - edges[0]

    bars = ax.bar(
        bin_centers, counts,
        width=bin_width * 0.85,
        color='steelblue',
        edgecolor='white',
        linewidth=0.8
    )

    # Etiquetas encima de cada barra
    for bar, count in zip(bars, counts):
        if count > 0:
            ax.text(
                bar.get_x() + bar.get_width() / 2,
                bar.get_height() + 0.3,
                str(count),
                ha='center', va='bottom', fontsize=9
```

```
            )

    ax.set_xlabel('F1 Score', fontsize=12)
    ax.set_ylabel('Cantidad de BSPs', fontsize=12)
    ax.set_title('Distribución de F1 Scores por BSP', fontsize=14)
    ax.set_xticks(bins)
    ax.set_xticklabels([f'{b:.1f}' for b in bins], rotation=45)
    ax.set_xlim(0, 1)

    # Línea de la media
    mean_f1 = best_f1s.mean()
    ax.axvline(mean_f1, color='red', linestyle='--', linewidth=1.5, label=f'Media: {mean_f1:.3f}')
    ax.legend(fontsize=10)

    plt.tight_layout()
    plt.show()

    print(f"Total BSPs: {len(best_f1s)}")
    print(f"Media F1: {mean_f1:.4f}")
    print(f"BSPs con F1 = 0: {(best_f1s == 0).sum()}")
    print(f"BSPs con F1 > 0.5: {(best_f1s > 0.5).sum()}")
    print(f"BSPs con F1 > 0.9: {(best_f1s > 0.9).sum()}")


# Ejecutar
plot_f1_distribution(best_f1s, bsp_pieces_names)
```

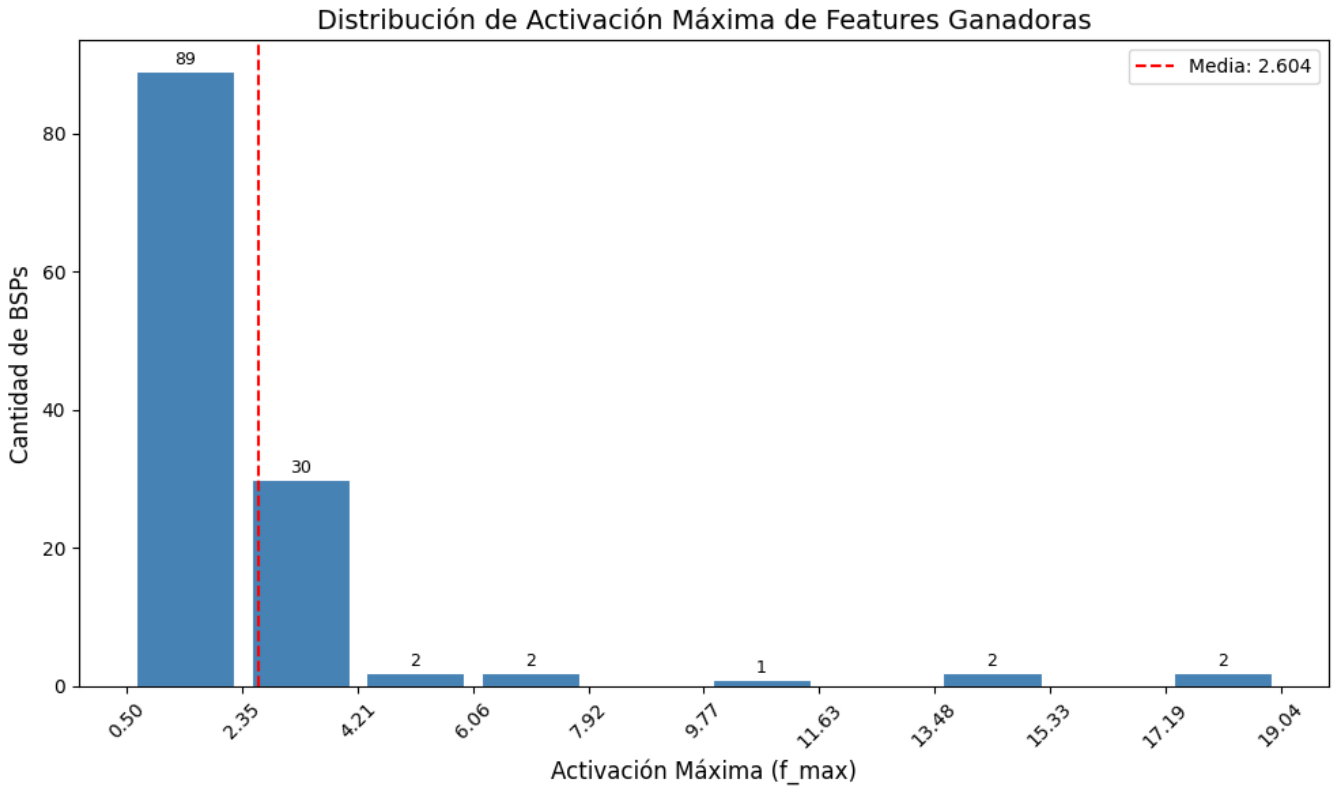



```
Total BSPs: 128
Media F1: 0.4623
BSPs con F1 = 0: 0
BSPs con F1 > 0.5: 38
BSPs con F1 > 0.9: 0
```

```python
# Extraer f_max de las features activas
f_max = sae_features.max(dim=0)[0]  # (n_features,)
```

```python
def plot_max_activation_distribution(best_features, f_max, bsp_pieces_names, n_bins=10):
    """
    Gráfico 2: Distribución de activación máxima de la feature ganadora por BSP.

    Args:
        best_features: array numpy con el índice de la feature ganadora de cada BSP
        f_max: tensor con la activación máxima de cada feature
        bsp_pieces_names: lista con los nombres de las BSPs
        n_bins: número de bins del histograma
    """
    # Obtener f_max de la feature ganadora de cada BSP
    # Ignorar BSPs sin feature ganadora (best_features == -1)
    valid_mask = best_features >= 0
    valid_features = best_features[valid_mask]
    f_max_winners = f_max[valid_features].cpu().numpy()

    fig, ax = plt.subplots(figsize=(10, 6))

    bins = np.linspace(f_max_winners.min(), f_max_winners.max(), n_bins + 1)
    counts, edges = np.histogram(f_max_winners, bins=bins)
    bin_centers = (edges[:-1] + edges[1:]) / 2
    bin_width = edges[1] - edges[0]

    bars = ax.bar(
        bin_centers, counts,
        width=bin_width * 0.85,
        color='steelblue',
        edgecolor='white',
        linewidth=0.8
    )

    # Etiquetas encima de cada barra
    for bar, count in zip(bars, counts):
        if count > 0:
            ax.text(
                bar.get_x() + bar.get_width() / 2,
                bar.get_height() + 0.3,
                str(count),
                ha='center', va='bottom', fontsize=9
            )

    # Línea de la media
    mean_fmax = f_max_winners.mean()
    ax.axvline(mean_fmax, color='red', linestyle='--', linewidth=1.5,
               label=f'Media: {mean_fmax:.3f}')
    ax.legend(fontsize=10)

    ax.set_xlabel('Activación Máxima (f_max)', fontsize=12)
    ax.set_ylabel('Cantidad de BSPs', fontsize=12)
    ax.set_title('Distribución de Activación Máxima de Features Ganadoras', fontsize=14)
    ax.set_xticks(edges)
    ax.set_xticklabels([f'{b:.2f}' for b in edges], rotation=45)

    plt.tight_layout()
    plt.show()

    print(f"Total BSPs con feature ganadora: {valid_mask.sum()}")
    print(f"BSPs sin feature ganadora: {(~valid_mask).sum()}")
    print(f"Media f_max ganadora: {mean_fmax:.4f}")
```

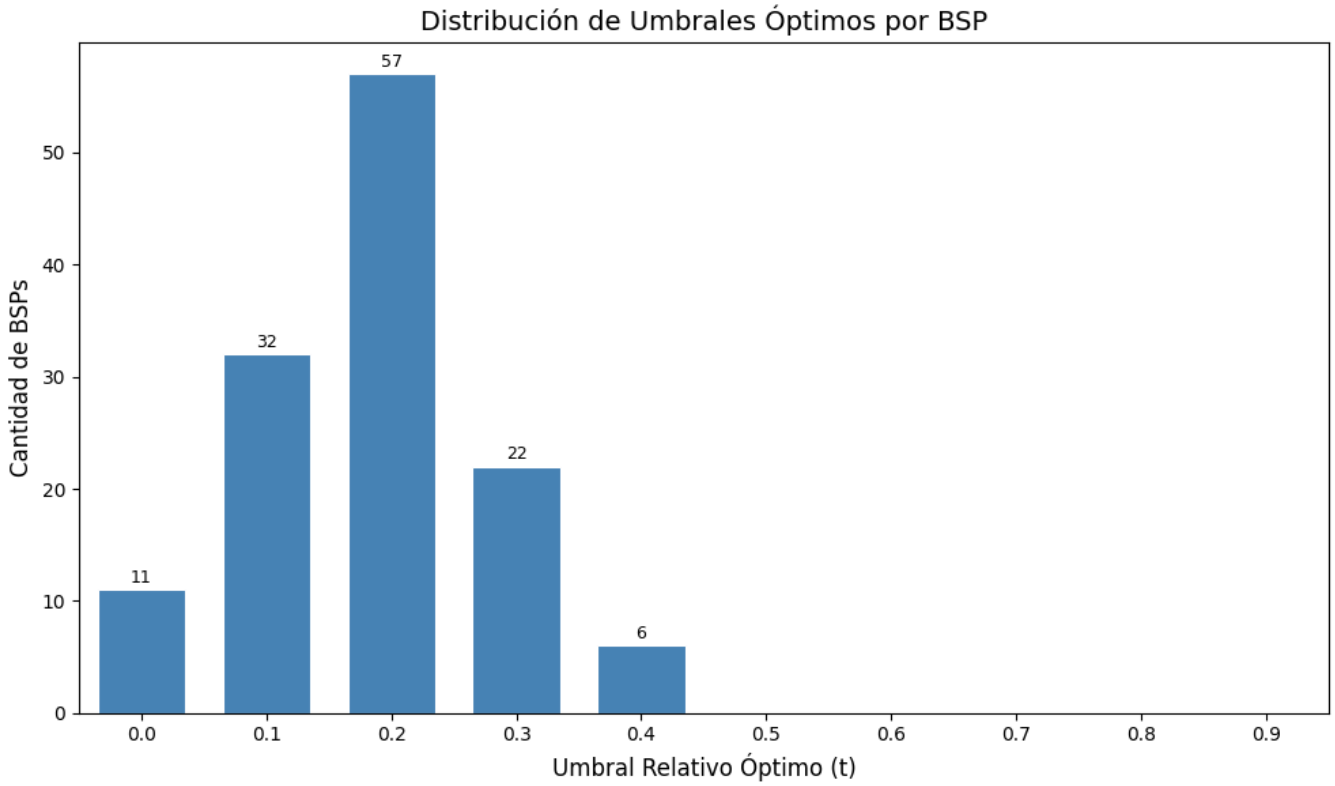
```
    print(f"Mínimo f_max: {f_max_winners.min():.4f}")
    print(f"Máximo f_max: {f_max_winners.max():.4f}")


# Ejecutar
plot_max_activation_distribution(best_features, f_max, bsp_pieces_names)
```




```
Total BSPs con feature ganadora: 128
BSPs sin feature ganadora: 0
Media f_max ganadora: 2.6037
Mínimo f_max: 0.4987
Máximo f_max: 19.0426
```

```
def plot_threshold_distribution(best_thresholds, bsp_pieces_names):
    """
    Gráfico 3: Distribución de umbrales óptimos por BSP.

    Args:
        best_thresholds: array numpy con el umbral óptimo de cada BSP
        bsp_pieces_names: lista con los nombres de las BSPs
    """
    # Ignorar BSPs sin feature ganadora (best_thresholds == -1)
    valid_mask = best_thresholds >= 0
    valid_thresholds = best_thresholds[valid_mask]

    fig, ax = plt.subplots(figsize=(10, 6))

    # Los umbrales son discretos: 0.0, 0.1, 0.2, ..., 0.9
    threshold_values = np.arange(0.0, 1.0, 0.1)
    counts = [(np.abs(valid_thresholds - t) < 0.05).sum() for t in threshold_values]

    bars = ax.bar(
        threshold_values, counts,
```

```
        width=0.07,
        color='steelblue',
        edgecolor='white',
        linewidth=0.8
    )

    # Etiquetas encima de cada barra
    for bar, count in zip(bars, counts):
        if count > 0:
            ax.text(
                bar.get_x() + bar.get_width() / 2,
                bar.get_height() + 0.3,
                str(count),
                ha='center', va='bottom', fontsize=9
            )

    ax.set_xlabel('Umbral Relativo Óptimo (t)', fontsize=12)
    ax.set_ylabel('Cantidad de BSPs', fontsize=12)
    ax.set_title('Distribución de Umbrales Óptimos por BSP', fontsize=14)
    ax.set_xticks(threshold_values)
    ax.set_xticklabels([f'{t:.1f}' for t in threshold_values])
    ax.set_xlim(-0.05, 0.95)

    plt.tight_layout()
    plt.show()

    print(f"Total BSPs con umbral óptimo: {valid_mask.sum()}")
    print(f"BSPs sin feature ganadora: {(~valid_mask).sum()}")
    print(f"Umbral más frecuente: {threshold_values[np.argmax(counts)]:.1f}")


# Ejecutar
plot_threshold_distribution(best_thresholds, bsp_pieces_names)
```



```
Total BSPs con umbral óptimo: 128
BSPs sin feature ganadora: 0
Umbral más frecuente: 0.2
```

0.8 8. Guardar Resultados

```
# Guardar resultados en el directorio de métricas del experimento
results = {
    'coverage': coverage,
    'best_f1s': best_f1s,
    'best_features': best_features,
    'best_thresholds': best_thresholds,
    'bsp_names': bsp_pieces_names,
    'execution_time_seconds': elapsed_time
}

metrics_dir = get_metrics_dir(
    NAMING_META['experiment_base_path'],
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos'],
    extras_id=extras_id
)
output_path = metrics_dir / f"coverage_{filter_suffix}.npz"
np.savez(output_path, **results)

print(f" Resultados guardados en:")
print(f"  {output_path}")
```

```
 Resultados guardados en:
  c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sae_standard\sae_l1-decay_standard\500games\ex
```

0.9 9. Generar PDF con Quarto

```
import subprocess

# Guardar PDF en el mismo directorio que el .npz (metrics/)
pdf_dir = get_metrics_dir(
    NAMING_META['experiment_base_path'],
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos'],
    extras_id=extras_id
)

pdf_name = get_report_name(
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos'],
    f'coverage_{filter_suffix}',
    extras_id=extras_id
)

notebook_name = 'coverage_gpu_optimized.ipynb'
output_path = pdf_dir / pdf_name
```

```
print(f' Generando PDF con Quarto...')
print(f' Archivo de salida: {output_path}')

# Quarto no acepta rutas en --output, solo nombre de archivo
# Se usa --output-dir para el directorio y --output solo para el nombre
result = subprocess.run(
    f'quarto render {notebook_name} --to pdf --output-dir "{pdf_dir}" --output {pdf_name}',
    shell=True,
    capture_output=True,
    text=True
)

if result.returncode == 0:
    print(f' PDF generado exitosamente: {output_path}')
else:
    print(f' Error al generar PDF:')
    print(result.stderr)
```

```
 Generando PDF con Quarto...
 Archivo de salida: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sae_standard\sae_l1-
decay_standard\500games\ex6\metrics\sae_standard_l1-decay_l6_500g_ex6_coverage_absolute.pdf
 PDF generado exitosamente: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sae_standard\sae_l1-
decay_standard\500games\ex6\metrics\sae_standard_l1-decay_l6_500g_ex6_coverage_absolute.pdf
```